

Tensor computations

Finite mean angular condition number for identifiable tensor decomposition

Difficulty: challenging **Importance:** interesting to specialist**Status:** Open

Rating rationale: Challenging because generic uniqueness does not supply the integrability estimates needed near degenerate decompositions; specialist importance reflects the particular angular condition number and volume-based input model.

Context and notation

Fix $d \geq 3$, $n_j \geq 2$, and $r \geq 2$. Assume generic complex identifiability: outside a proper algebraic exceptional set, a complex tensor of rank r in this format has a unique unordered collection of r rank-one summands. Let M_r be the smooth identifiable locus of real rank- r tensors, with induced Euclidean volume dV . The random input has density

$$d\mu(A) = Z^{-1} e^{-\|A\|_F^2/2} dV(A).$$

For the addition map $\Phi(a_1, \dots, a_r) = \sum_i a_i$ on rank-one tensors, let Ψ be a local inverse at A . Use Frobenius norms and their product norm. Define

$$\kappa(A) = \|D\Psi(A)\|_2, \quad \kappa_{\text{ang}}(A) = \|D(p^{\times r} \circ \Psi)(A)\|_2, \quad p(a) = a/\|a\|_F.$$

Values on measure-zero exceptional sets do not affect the expectations. This samples tensors by volume, not independent Gaussian summands.

Problem statement

Under this probability model, prove or disprove

$$\mathbb{E}_\mu \kappa_{\text{ang}}(A) < \infty$$

for every admissible format and $r \geq 3$. The rank-two case is a theorem. Normalizing each recovered summand inside the derivative is part of the definition; simply dividing the regular condition number by a scalar does not give this problem.

Reference

Beltrán, Breiding, and Vannieuwenhoven, *The Average Condition Number of Most Tensor Rank Decomposition Problems Is Infinite*, *Foundations of Computational Mathematics* 23 (2023), 433–491: equation (6), Theorem 3, and Conjecture 2. The [author preprint](#) labels the conjecture 1.10.

Status check — 2026-09-10

Rechecked the [journal version](#), [equation \(6\)](#), [Theorem 3](#) and [Conjecture 2](#), and searched for higher-rank angular-condition-number proofs. The theorem covers rank two, which the displayed target already excludes; no result proving finite mean for all admissible ranks at least three was located. The 2023 conjecture is the latest explicit statement checked, and a later unlocated result cannot be ruled out.